

PITTSBURGH INTL, PA, USA

WMO: 725200

Lat: 40.5317N

Lon: 80.2172W

Elev: 367

StdP: 97.00

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 94823

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.4	-12.7	-20.2	0.6	-13.6	-18.0	0.8	-11.3	11.3	-1.3	10.5	-2.8	3.2	260	0.486

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	31.6	22.2	30.2	21.6	29.0	20.9	23.9	29.0	23.1	27.7	22.4	26.8	4.0	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.4	17.8	26.3	21.6	17.0	25.5	20.9	16.3	24.7	73.2	29.2	70.1	27.8	67.3	26.8	27.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)		
(4) 9.9	8.5	7.6		-18.6	33.2	3.5	1.7	-21.1	34.5	-23.2	35.5	-25.2	36.5	-27.8		
(5)				-19.0	25.3	3.5	1.0	-21.5	26.0	-23.5	26.6	-25.5	27.1	-28.0	27.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.2	-1.9	-0.3	4.4	11.1	16.4	20.9	22.8	22.2	18.5	12.1	6.0	1.1	(6)
(7)		DBStd	10.02	6.84	6.31	6.24	5.33	4.51	3.41	2.73	2.72	3.80	4.78	5.20	5.61	(7)
(8)		HDD10.0	1371	373	293	197	50	4	0	0	0	0	32	141	280	(8)
(9)		HDD18.3	3066	627	522	434	223	94	15	2	3	43	200	370	534	(9)
(10)		CDD10.0	1801	4	5	23	83	202	328	398	377	257	98	21	5	(10)
(11)		CDD18.3	453	0	0	1	6	32	93	142	122	49	8	0	0	(11)
(12)		CDH23.3	3605	0	1	7	68	303	742	1131	924	378	49	1	0	(12)
(13)		CDH26.7	1001	0	0	0	10	66	213	345	263	97	7	0	0	(13)
(14)	Wind	WSAvg	3.3	4.0	3.9	4.0	4.0	3.3	3.0	2.6	2.5	2.6	3.0	3.4	3.6	(14)
(15)	Precipitation	PrecAvg	959	69	62	83	80	95	100	99	84	80	63	73	72	(15)
(16)		PrecMax	1468	159	179	162	140	167	261	221	200	256	123	281	216	(16)
(17)		PrecMin	656	20	14	24	12	33	16	18	25	7	4	10	13	(17)
(18)		PrecStd	157	31	31	33	31	38	49	43	44	46	28	41	32	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.0	19.3	24.4	28.2	30.4	32.5	33.4	33.1	31.7	27.4	21.9	18.0	(19)
MCWB			13.3	12.5	14.9	16.9	20.6	22.4	23.3	22.8	21.1	19.7	15.1	13.7	(20)	
2%		DB	14.5	15.7	21.0	25.3	28.5	30.5	31.5	31.1	29.5	24.5	19.0	15.0	(21)	
		MCWB	11.9	11.0	13.2	15.1	19.2	21.8	22.6	22.0	20.8	17.7	13.1	11.9	(22)	
5%		DB	11.6	12.4	17.8	22.9	26.9	29.2	30.0	29.5	27.3	22.5	16.8	12.4	(23)	
		MCWB	9.2	8.4	11.4	14.4	18.6	21.1	21.8	21.4	19.9	16.5	12.1	9.8	(24)	
10%		DB	7.9	9.0	14.5	20.3	24.8	27.6	28.6	28.0	25.5	20.2	14.6	9.7	(25)	
		MCWB	5.5	5.7	9.8	12.9	17.7	20.4	21.1	20.6	19.1	15.7	10.9	7.2	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.4	14.4	16.0	18.9	22.1	24.1	25.2	24.6	23.7	21.4	16.8	14.8	(27)
MCDWB			16.8	18.2	22.4	25.3	28.4	30.0	30.8	30.0	28.7	25.4	19.7	17.0	(28)	
2%		WB	12.4	11.9	14.3	16.9	20.8	23.1	24.1	23.6	22.3	19.6	14.7	12.4	(29)	
		MCDWB	14.4	14.6	18.8	22.3	26.3	28.4	29.1	28.3	26.7	22.6	17.8	14.1	(30)	
5%		WB	9.5	8.7	12.6	15.7	19.8	22.3	23.2	22.8	21.2	17.9	12.9	10.2	(31)	
		MCDWB	11.3	12.0	16.9	20.8	24.9	27.3	27.7	27.1	25.1	21.1	16.0	12.0	(32)	
10%		WB	5.7	5.9	10.2	14.3	18.7	21.4	22.4	21.9	20.3	16.2	11.1	7.2	(33)	
		MCDWB	7.6	8.9	14.1	18.8	23.3	26.0	26.6	26.0	23.9	19.8	14.1	9.4	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.9	8.7	9.9	11.5	11.0	10.2	10.0	10.4	10.0	8.8	7.3	(35)	
MCDBR			10.6	13.3	14.3	14.5	13.3	11.9	11.3	11.6	12.3	12.8	12.2	10.9	(36)	
MCWBR			8.6	10.0	8.5	7.6	6.2	5.1	4.4	4.6	5.2	6.8	8.2	8.8	(37)	
5% WB		MCDBR	10.0	12.3	12.3	12.3	11.2	10.3	9.6	9.5	9.7	10.3	10.6	10.3	(38)	
		MCWBR	8.8	10.1	8.3	7.5	5.8	4.9	4.5	4.6	4.7	6.4	8.2	9.3	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.302	0.326	0.352	0.380	0.421	0.452	0.457	0.454	0.387	0.362	0.318	0.300	(40)	
taud		2.461	2.374	2.380	2.331	2.227	2.177	2.179	2.175	2.339	2.412	2.514	2.512	(41)		
Ebn at Noon		861	886	899	893	859	830	821	812	849	833	837	837	(42)		
Edh at Noon		78	98	109	122	139	147	145	142	112	94	74	69	(43)		
(44)	All-Sky Solar Radiation	RadAvg	1.50	2.36	3.31	4.51	5.27	5.71	5.76	5.21	4.25	2.83	1.91	1.27	(44)	
(45)		RadStd	0.14	0.34	0.32	0.45	0.45	0.47	0.40	0.30	0.42	0.27	0.26	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.34	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+101	N/A
(47) Regional (40 neighbors)		+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+125	+59

Nomenclature: See separate page